Graphs and Networks - Final Assignment

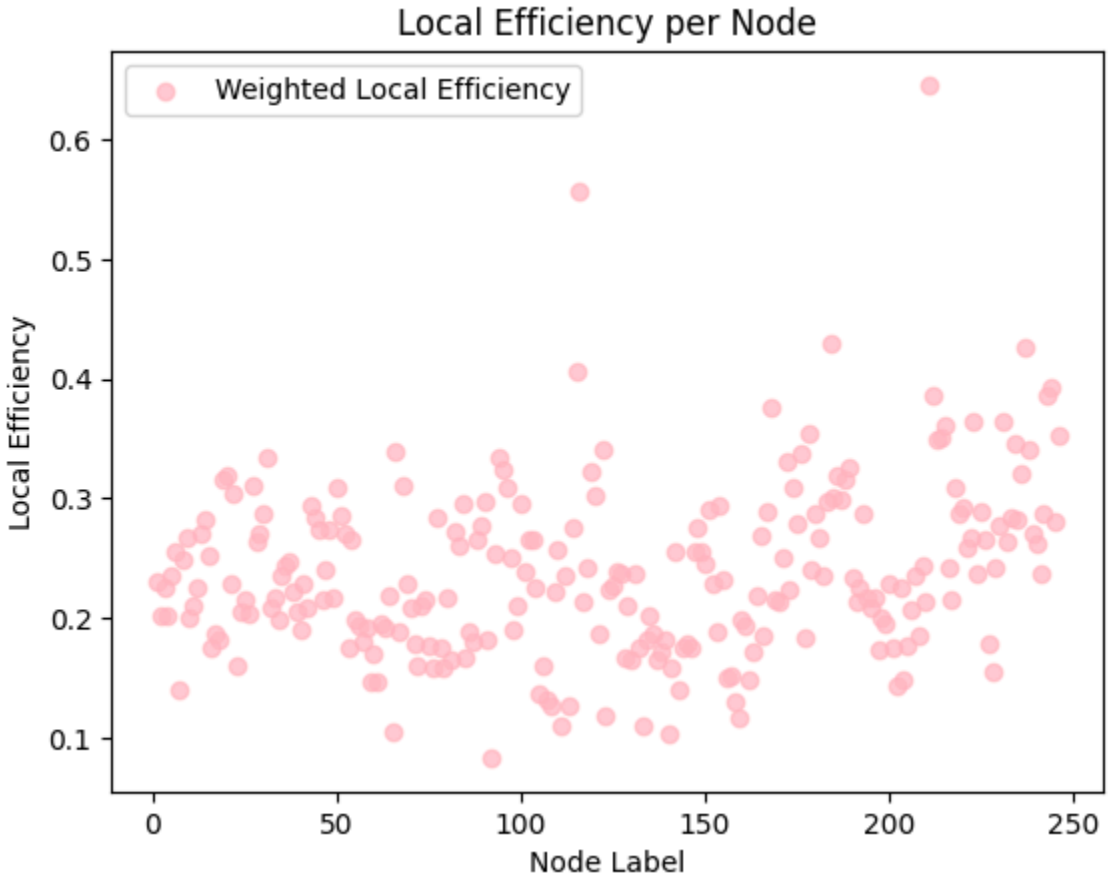
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Chosen Dataset: Structural

Assumption: structural data values are the amount of streamlines between the nodes.

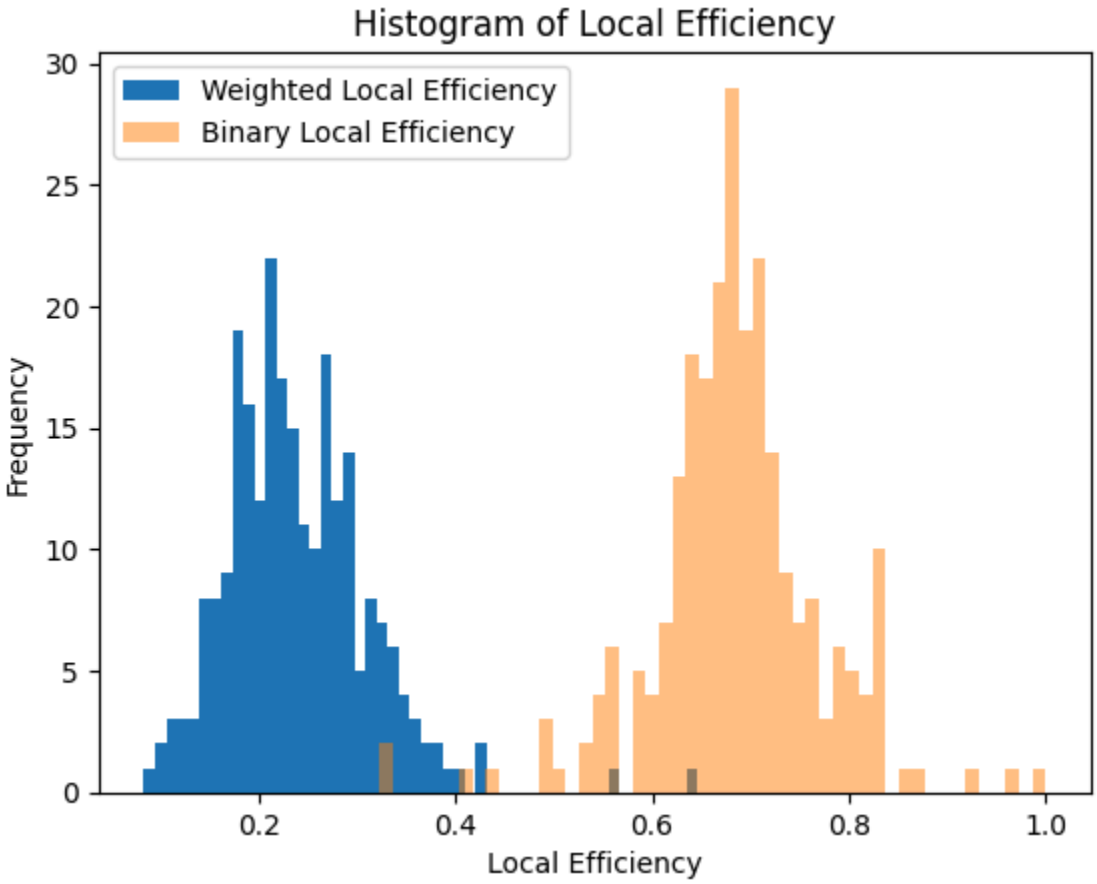
Question 1

1. Local efficiency measures how well information is exchanged within the immediate neighborhood of a node, quantifying the efficiency of communication between a node's direct connections when the central node is removed.
2. Plot:



Caption: Local Efficiency of each node in the network, helps identify which nodes have extreme efficiency values. In addition, can help easily identify the local efficiency of a specific node, and compare to other nodes in the network.

3. We used weighted edges. If I use binary edges, we'll lose the dimension of strength of the connection, and the network will look more efficient than in weighted.



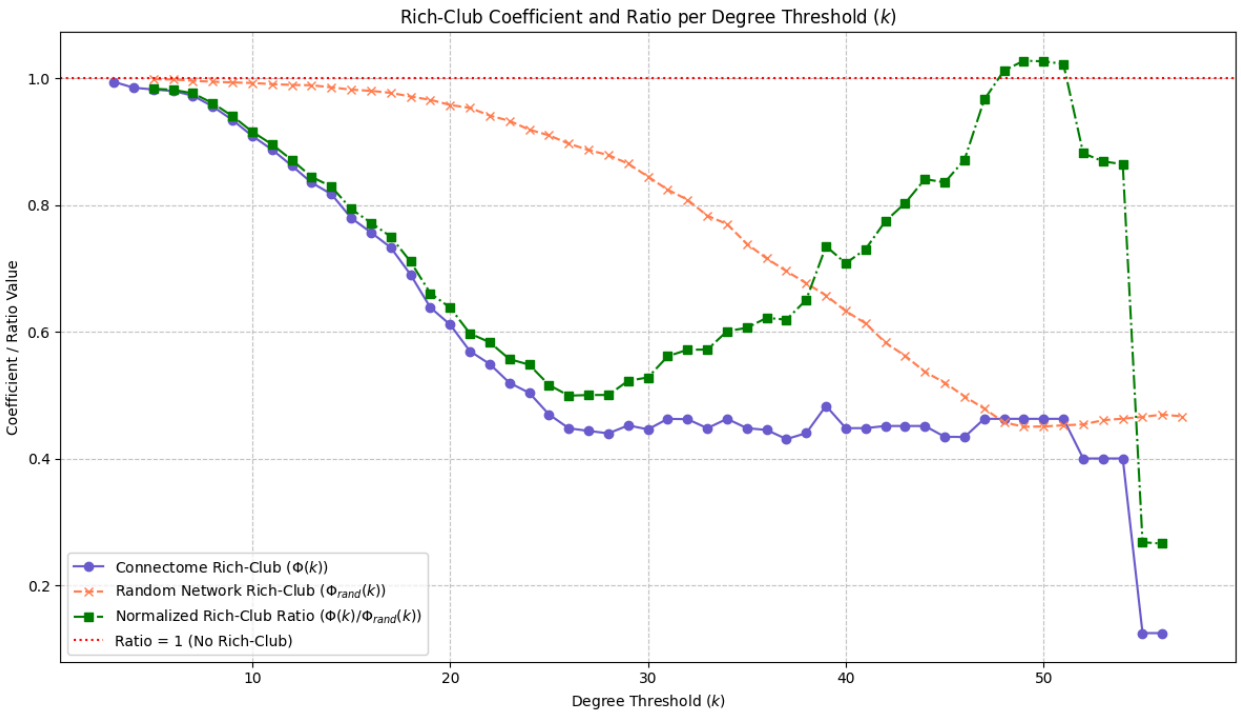
4. When considering a brain network, using weighted edges generally provides better results than binary edges, especially for structural connectomes.
    a. Pros of using Weighted Edges:
        - Richer Information Content: Weighted edges capture the strength of connections. This provides a more nuanced and biologically realistic representation of brain connectivity than simply knowing if a connection is present or absent. Stronger connections are likely to facilitate more efficient information flow, and weighted metrics account for this.
        - Increased Sensitivity: By incorporating weight information, weighted graph measures can be more sensitive to subtle changes in brain network

organization that might be missed by binary analyses. This is particularly important in clinical studies where small differences in connectivity might be indicative of disease states or treatment effects

b. Cons of using Weighted Edges**:**
   - Interpretation Complexity: Efficiency values require careful interpretation and understanding of the underlying normalization conventions (e.g., BCT's 1/Wij transformation). This can make direct comparison across studies using different software or normalization approaches more challenging.
   - Sensitivity to Noise/Artifacts: The magnitude of weights can be influenced by noise or artifacts in the MRI data acquisition and processing. Very high or very low weights, if not truly reflective of biological connectivity, can disproportionately impact weighted graph measures, potentially leading to spurious results. Binary networks, while losing information, can sometimes be more robust to certain types of noise if an appropriate threshold is applied.

Question 2

1. The rich-club coefficient $\phi(k)$ measures the tendency of high-degree nodes (degree $\geq k$) to be densely interconnected. It's computed as the ratio of existing to possible edges within this group of nodes with degree $\geq k$. A high $\phi(k)$ indicates a "rich club": a tightly inter-connected core of central nodes, suggesting a hierarchical organization and efficient network backbone.
2. Plot:



3. The range of k values in which the nodes are rich club nodes is found by comparing the rich-club coefficient for each possible k of the given network to a random network (or a set of random networks) with similar features. The range of k is found by the comparison of the curve, computing whether the coefficient is higher than in a random network (the ratio is more than 1). In our case, the rich-club coefficient slightly exceeds 1 at the range of $48 \leq k \leq 51$, so nodes of these degrees show rich-club values. Since the calculation is based on randomized networks, results may vary slightly from run to run.

Question 3

1. One can say that a network has a small-world topology if it exhibits two characteristics simultaneously: a high clustering and a low average path length. Such network structure is believed to be the most balanced from cost and efficiency perspective.
   An exemplary small-world network can be obtained via Watts-Strogatz model, i.e. from a regular lattice network by an iterative rearrangement of its edges from given nodes *i,j* to some random *i,h* | *h≠j* with a given probability *p*, where *p*=0 is no changes made, *p*=1 is all the edges have been shuffled. During this rearrangement network will be transformed from regular (having both high clustering and high average path length) into random (both low clustering and low path length). However, when implementing Watts-Strogatz

algorithm, network properties tend to change non-linearly with each iteration: average path length decreases rapidly already at the early steps (displaying random-like topology), while high clustering persists for more steps (regular-like topology). This middle-state is referred to as the small-world regime.

2. Small-worldness is measured by comparison of a given network's average path length and clustering to some reference network. Depending on whether we take a regular or a random (usually, Erdos-Renyi) network as a reference, small-worldness measures are sigma and omega respectively.

   Sigma is defined as relation of given network's clustering coefficient Cl to average Cl of a set of random networks of similar properties divided by relation of network's average path length L to average L of a set of random networks:

   $$\sigma = \frac{Cl/\langle Cl\rangle}{L/\langle L\rangle}$$

   Omega is defined in the same manner, but with regular network's clustering coefficient as a reference:

   $$\omega = \frac{\langle L_{rand}\rangle}{L} - \frac{Cl}{\langle Cl_{reg}\rangle}$$

   For the given structural connectome measures' values are:

   Sigma: 3.65

   Omega: 0.37

   Furthermore, it is possible to compute weighted sigma and omega indices (though usually small-worldness is assessed on binary graphs) by computing weighted average path length and weighted average clustering (e.g. with the formula proposed by Onnela et al. (2005)). In this case, measures for the network will be:

   Weighted Sigma: 0.71

   Weighted Omega: 0.44

3. We can assess to which degree a network exhibits small-world topology by sigma or omega values. Thus, if the sigma value is greater than 1, the network is considered small-world.

   $$\sigma = \frac{Cl/\langle Cl\rangle}{L/\langle L\rangle}$$

   In turn, negative values of omega mean the network's structure is simi
   lar to a regular one, while positive – to a random one. Values close to 0 mean that the network has small-world characteristics.

As we can see, our connectome displays a fair small world topology. Sigma value stands at 3.65, which is significantly greater than1, and omega equals 0.37, which suggests a trend towards random topology, but is still closer to 0, than to 1, so the net can be called small-world.

Worth noting is that both measures when weighted suggest lower small-worldness, which is inline with our findings on local efficiency variation in weighted and unweighted networks.

Question 4

1. For **segregation** we chose - clustering coefficient:

   Clustering coefficient is a measure (for each node) that represents local interconnections of neighbors of node i.

   The clustering coefficient for each node is defined as the ratio between the number of pairs of its neighbors that are connected to each other, and the total number of possible pairs of neighbors for that node. In this case we used the common approach of non-weighted edges, to reduce complexity.

   For **integration** we chose – global efficiency:

   Global efficiency is the multiplicative inverse of the mean shortest path.

   So in other words, it measures the inverse of the average shortest path length in the network, quantifying how efficiently information is exchanged across all nodes. In this measure we used weighted edges, as it's more informative and simple to use.

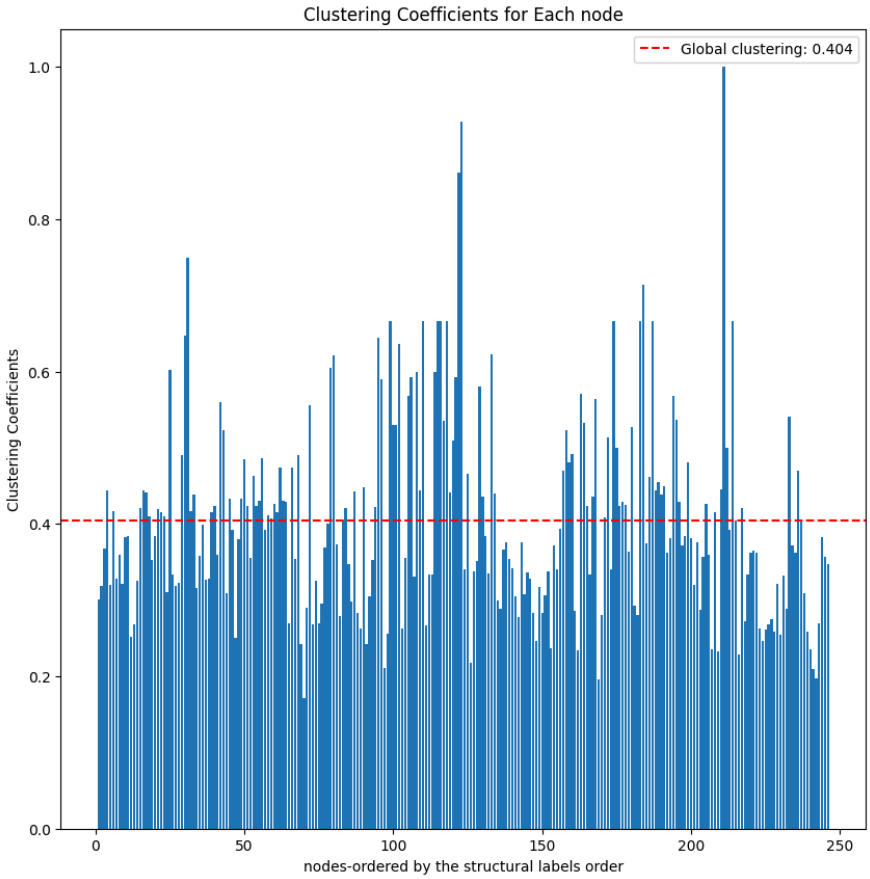


Fig 1: Node-wise clustering coefficients with global clustering coefficient (mean) as dashed red line.

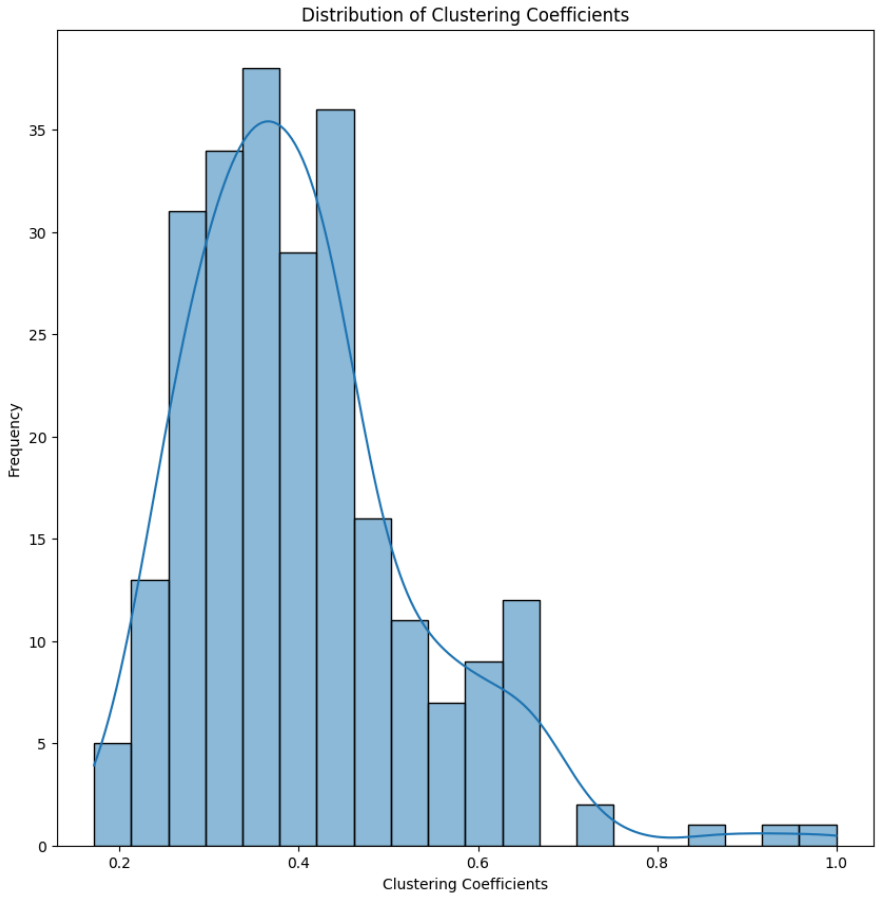


Fig 2:distribution of clustering coefficients.

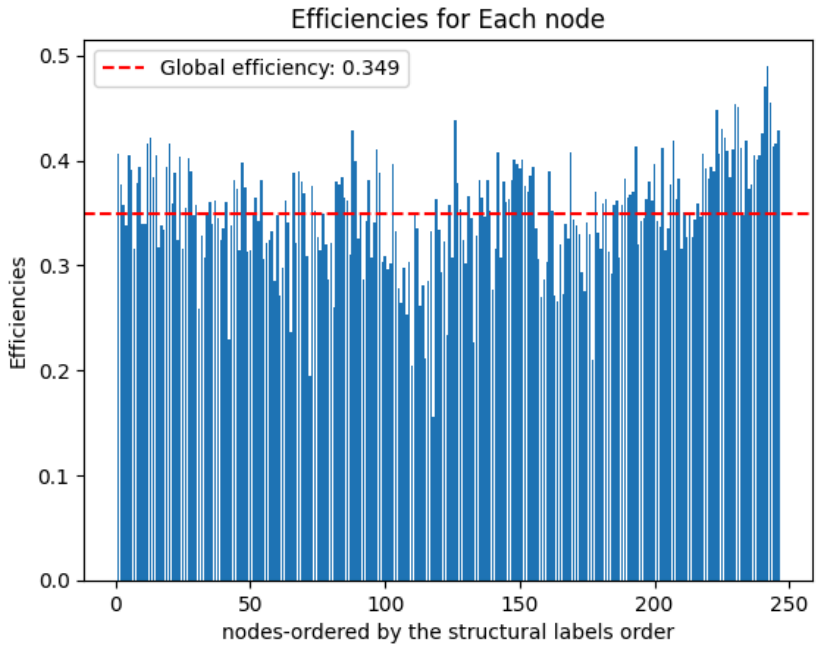


Fig 3: bar graph that represents each nodes' efficiency with the global efficiency as a red dotted line.

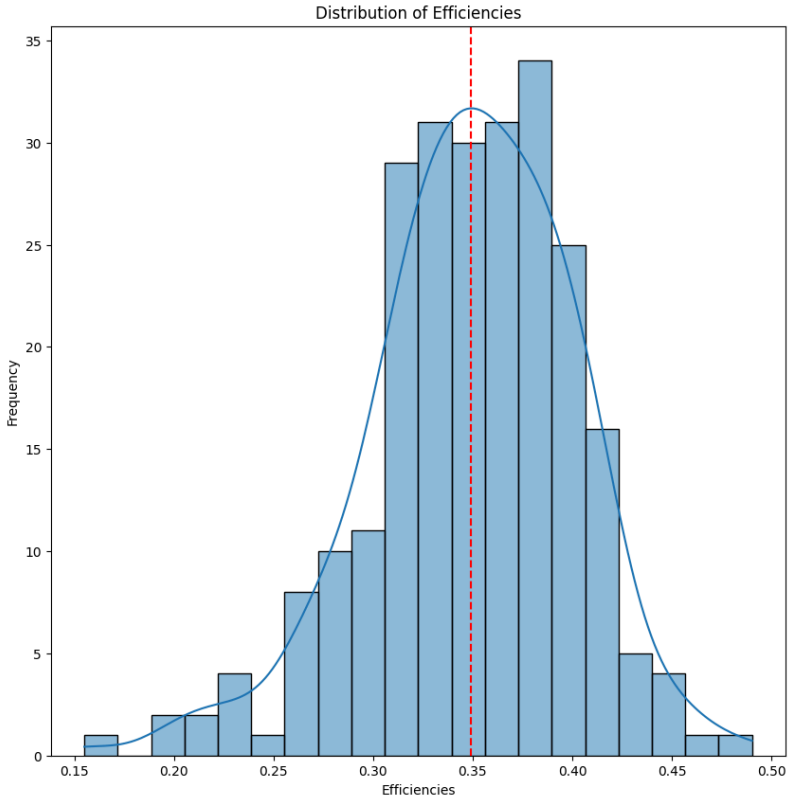


Fig 4: distribution of all efficiencies with the global efficiency as a red dotted line.

Based on the graphs, the clustering coefficients exhibit considerable variability across nodes, reflecting heterogeneity in local interconnectivity. The global clustering coefficient of 0.404 indicates that approximately 40% of all possible neighbor pairs are interconnected, suggesting a tendency for nodes to form locally clustered neighborhoods.

In terms of integration, the global efficiency of approximately 0.35 indicates that, on average, information exchange between pairs of nodes occurs at about 35% of the efficiency of a fully connected network. This suggests that the network supports moderately efficient communication across distant regions.

Overall, the network displays substantial local clustering alongside a moderate level of global integration, reflecting a balance between localized specialization and distributed information transfer.

Question 5

1. We removed the node with highest degree centrality value (taking into account the weights)

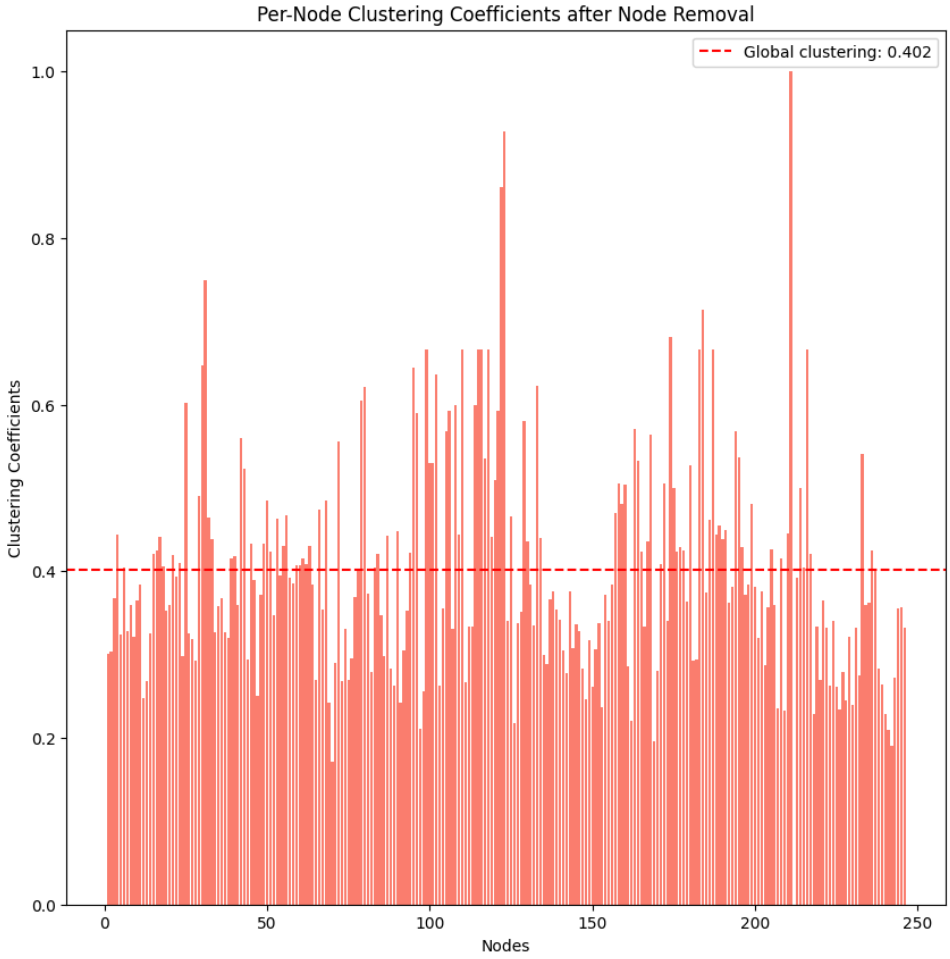


Fig 5: Bar graph for each nodes' clustering coefficients after central node removal, with a dashed line indicating average clustering.

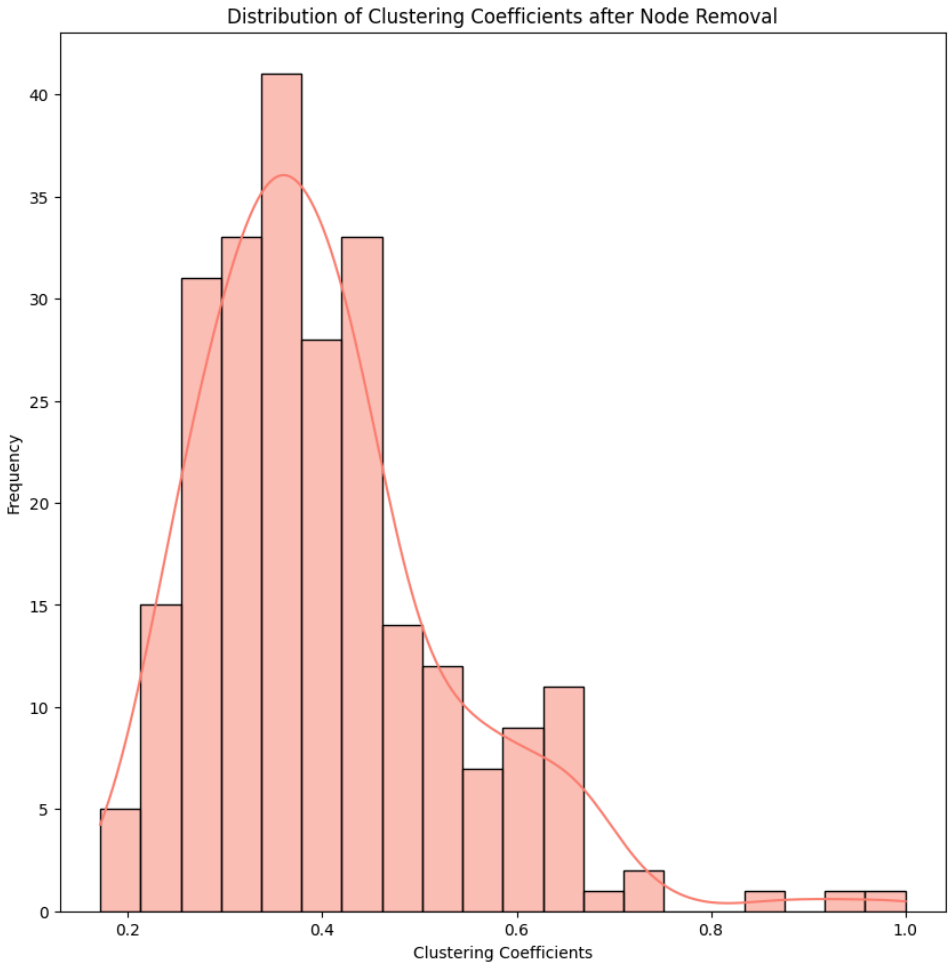


Fig 6: Distribution of clustering coefficients of remaining nodes after central node removal.

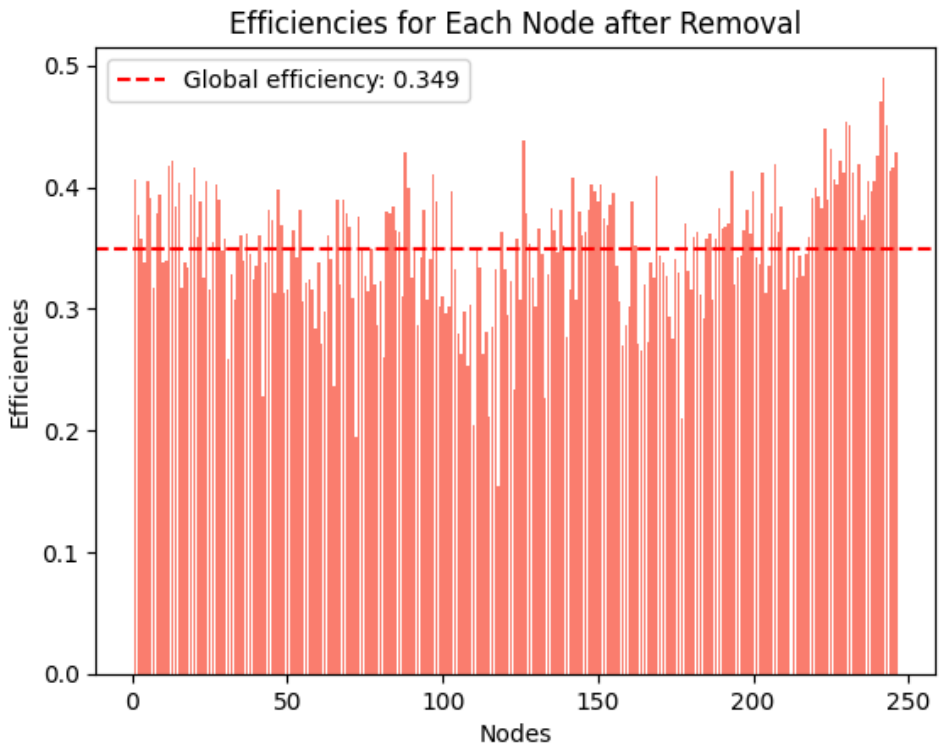


Fig 7: Bar graph for each nodes' efficiency after removal of the central node, with the global efficiency represented as a red dotted line.

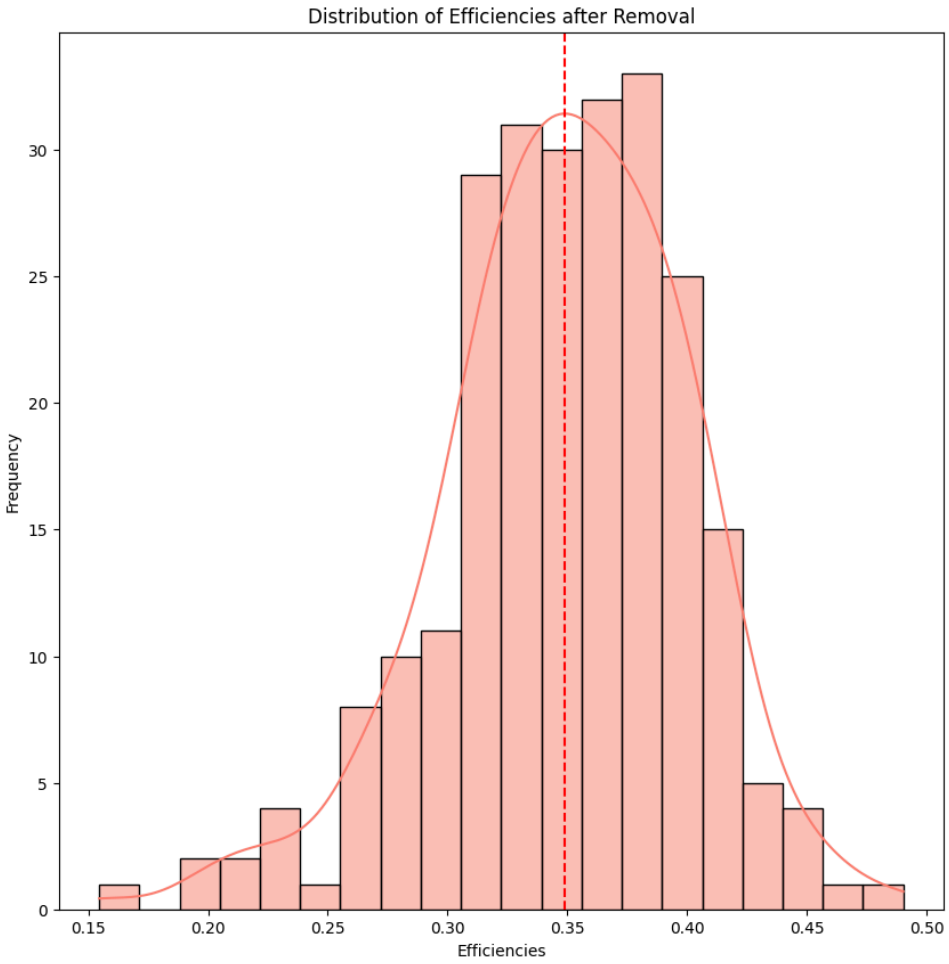


Fig 8: Distribution of nodal efficiencies, after removal of the central node, with the global efficiency represented as a red dotted line .

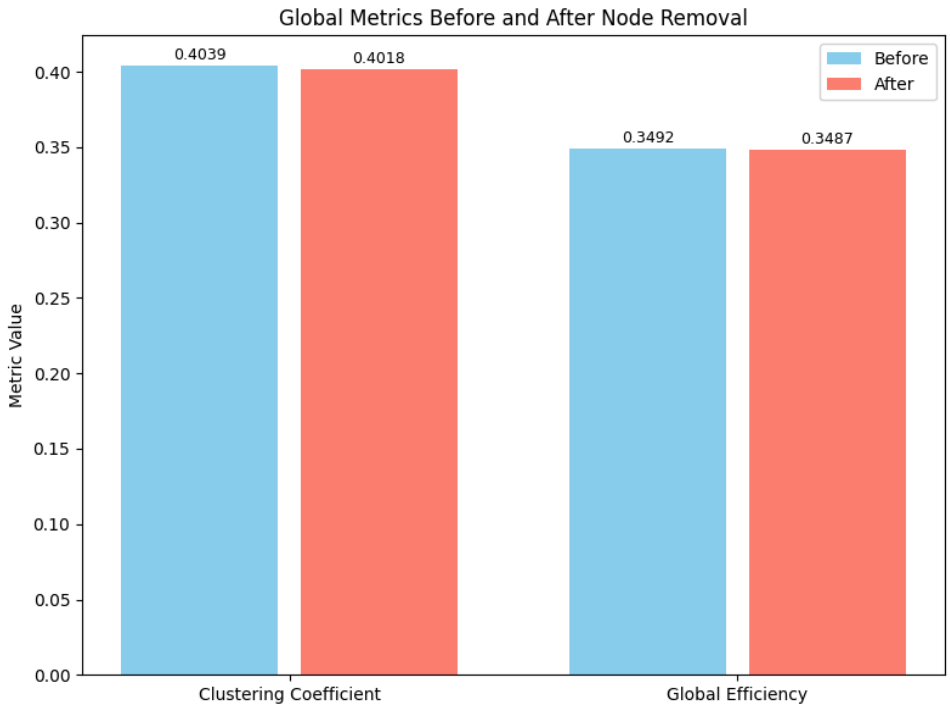


Fig 9: Comparison of global clustering coefficient and global efficiency, before and after removal of a central node.

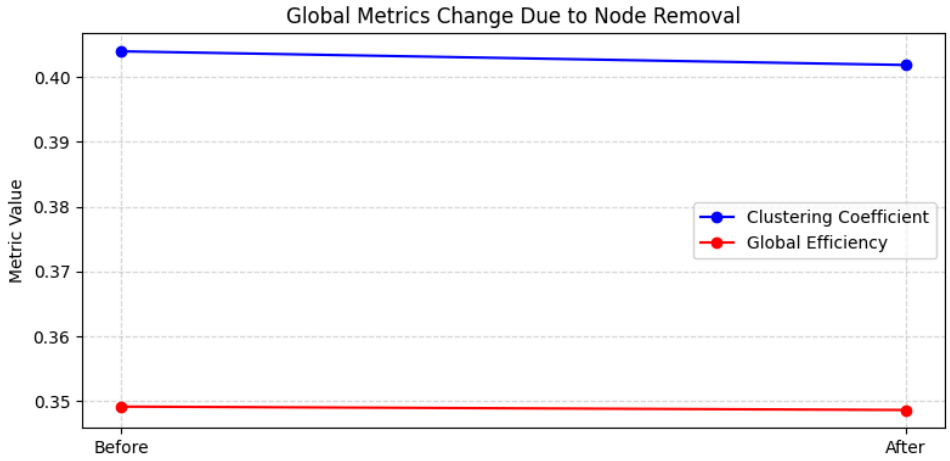


Fig 10: Comparison of global clustering coefficient and global efficiency as line plots, before and after removal

2. We can see miniscule changes in values (for example a decrease in mean clustering coefficient of 0.002 or decrease of 0.0005 in global efficiency). This minimal impact reflects the robustness of the structural connectome, where multiple pathways support efficient communication even after node loss.